

AIGR Gene Hypothesis Deep Research —

S. pombe Dca7 (O74763)

Gene: dca7 (SPBC17D11.08) · **UniProt:** O74763 · **Organism:** *Schizosaccharomyces pombe* (972 / ATCC 24843) **Focus type:** core_function **Seed hypothesis:** *S. pombe* Dca7 is a bona fide substrate receptor component of a Cul4-Ddb1 RING ubiquitin ligase complex.

Summary

Verdict: Weakly supported / over-annotated. The claim that *S. pombe* Dca7 (UniProt O74763; systematic name SPBC17D11.08) is a bona fide substrate-receptor subunit of a Cul4-Ddb1 (CRL4) RING ubiquitin ligase rests entirely on orthology transfer and a historically loaded protein name ("DCAF7" = **DDB1-** and **CUL4-Associated Factor 7**). It is **not** supported by any direct experimental evidence in fission yeast. The single cellular-component GO annotation placing Dca7 in a Cul4-RING E3 ubiquitin ligase complex (GO:0080008) carries an ISS (Inferred from Sequence/Structural Similarity) evidence code with GO_REF:0000024 — i.e., a manual transfer of annotation to an ortholog, not an experimental observation in *S. pombe*.

Three independent lines of analysis argue that the CRL4 substrate-receptor role is **not** the primary (core) function of Dca7. First, Dca7 lacks the conserved **WDxR** DDB1-binding motif that genuine WD40-type DCAFs use to dock onto the DDB1 β -propeller; a motif scan of the 435-residue sequence returned zero WDxR matches. Second, PomBase records no physical or genetic interaction between Dca7 and any *S. pombe* CRL4 core subunit (Pcu4/Cul4, Ddb1, Rik1, Cdt2); its only two curated physical interactors are the chromatin remodeller Fft3 and, tellingly, the dual-specificity protein kinase Ppk15. Third, the deeply conserved and experimentally validated function of this entire WD40 β -propeller family — human DCAF7/WDR68/HAN11, petunia AN11, *C. elegans* Swan-1/2 — is that of a **catalytically inactive scaffold** that binds and coordinates DYRK-family, HIPK, and MEKK kinases, regulating their stability, localization, and downstream gene-expression programs.

The competing "kinase scaffold" model is not merely a plausible alternative; it is the better-evidenced core function across the family, and it is echoed in fission yeast by Dca7's interaction with Ppk15, a kinase annotated with both Ser/Thr and Tyr kinase activities — the dual-

specificity signature of the DYRK family. In balance, one report does document that human WDR68 can physically associate with CUL4-DDB1, so the DCAF designation is not baseless; however, **no study in any organism demonstrates DCAF7/WDR68/Dca7 delivering a substrate for CUL4-mediated ubiquitination.** For curation, the Cul4-RING complex annotation on Dca7 should be treated as low-confidence/orthology-only, and the gene's core function is better captured by a kinase-scaffold / gene-expression regulatory role.

Executive Judgment

Weakly supported / over-annotated. *S. pombe* Dca7 (O74763) is the ortholog of human DCAF7/WDR68, and the sole support for it being a Cul4-Ddb1 substrate receptor is an orthology-transferred GO term (GO:0080008, ISS/GO_REF:0000024) inherited from the historically loaded "DCAF7" name. There is no direct *S. pombe* evidence, no WDxR DDB1-binding motif (0 matches), and no recorded interaction with any pombe CUL4 subunit (Pcu4/Ddb1/Rik1/Cdt2). The deeply conserved, experimentally validated core function of this WD40 β -propeller family is a **catalytically inactive scaffold for DYRK/HIPK/MEKK kinases** — echoed in fission yeast by Dca7's interaction with the dual-specificity kinase Ppk15. A single mammalian report shows WDR68 can associate with CUL4-DDB1, so the DCAF label is not baseless, but substrate delivery has never been demonstrated. Curators should treat the Cul4-RING annotation as low-confidence/orthology-only, not a demonstrated substrate-receptor core function.

Most important caveats: (1) "No recorded interaction" reflects curated PomBase datasets, which are not exhaustive; (2) some DCAFs dock onto DDB1 through non-WDxR surfaces, so motif absence is strong but not absolute proof; (3) the Ppk15 link rests on a single two-hybrid plus GO-level dual-specificity annotation.

Key Findings

Finding 1 — The Cul4-RING annotation on Dca7 is an orthology transfer (ISS), not experimental evidence

Dca7 (O74763 / SPBC17D11.08) is the *S. pombe* ortholog of human **DCAF7/WDR68**. UniProt places it in InterPro family **IPR045159 "DCAF7-like"**, and PomBase lists the human ortholog as HGNC:30915 (DCAF7/WDR68), the *S. cerevisiae* ortholog as YPL247C, and the *S. japonicus* ortholog as SJAG_04087.

Critically, the only cellular-component GO annotations on Dca7 are:

- **GO:0080008** "Cul4-RING E3 ubiquitin ligase complex" — evidence code **ISS**, reference **GO_REF:0000024** (manual transfer of experimentally-supported annotations to orthologs), and
- **GO:0005634** "nucleus" — evidence code **IBA** (Inferred from Biological ancestor), reference **GO_REF:0000033**.

There is **no molecular-function and no biological-process GO term** on the gene, and no direct assay placing Dca7 in a CRL4 complex. The 28 references associated with the gene in PomBase are high-throughput proteomics/interactome/genetic-screen datasets — none is a dedicated CRL4 biochemistry paper focused on Dca7. The biochemical criterion that a WD40 protein must meet to be a genuine Cul4-Ddb1 substrate receptor was established by [PMID: 16949367](#): "*DCAFs interact with multiple surfaces on Ddb1, and the interaction of WD40-containing DCAFs with Ddb1 requires a conserved 'WDXR' motif.*" This provides the objective standard against which Dca7 can be judged — and Dca7 fails it (Finding 2).

Finding 2 — Dca7 lacks a WDxR DDB1-binding motif and shows no interaction with the *S. pombe* CRL4 core

A motif scan of the 435-amino-acid O74763 sequence found **zero WD-x-R (WDxR) matches** (only two bare "WD" dipeptides, at positions 194 and 343). The conserved DDB1-binding box required of WD40-type DCAFs is therefore absent from Dca7. This is a direct, sequence-level discriminator: the very feature [PMID: 16949367](#) defines as necessary for a WD40 DCAF to bind Ddb1 is missing.

Consistent with the absence of a docking motif, PomBase records only **two physical interactors** for Dca7, neither of them a CRL4 subunit:

- **Fft3** (SPAC25A8.01c), a SMARCAD1-family chromatin remodeller (Affinity Capture-MS, [PMID: 28218250](#)), and
- **Ppk15** (SPAC823.03), a Ser/Thr protein kinase (two-hybrid, [PMID: 26771498](#)).

No physical or genetic interaction is recorded with *S. pombe* Cul4/Pcu4, Ddb1, Rik1, Cdt2, or any other CRL4 component. PomBase's curated product description reads: "*WD repeat protein, DDB1 and CUL4-associated factor Dca7, implicated in gene expression*"; the deletion is viable. Both the interactor profile (a kinase and a chromatin remodeller) and the "implicated in gene expression" descriptor fit the DCAF7 kinase-scaffold / gene-regulation paradigm far better than a CRL4 substrate-receptor role. The competing paradigm is captured by [PMID: 23349862](#): the vertebrate ortholog "*Wdr68 is a highly conserved scaffolding protein... A Ras-Map3k-Wdr68-Dyrk1 signaling relay may mediate these and other diverse signaling events.*"

Finding 3 — The conserved core function of the DCAF7/WDR68/Han11 family is a catalytically inactive kinase scaffold

Across eukaryotes, the Dca7 ortholog is characterized as a scaffold/adaptor with **no catalytic activity** that binds and coordinates protein kinases:

- Han11 directly binds HIPK2, DYRK1A, DYRK1B and MEKK1 and is required to couple MEKK1 to DYRK1/HIPK2 ([PMID: 20940704](#)): "*these experiments identify Han11 as a novel scaffold protein regulating kinase signalling by HIPK2 and MEKK1.*"
- DCAF7 "*is an evolutionarily conserved protein with a single WD40 repeat domain and has no catalytic activity*" and acts as a substrate-recruiting subunit of DYRK1A and HIPK2 ([PMID: 27307198](#)).
- WDR68 binds the N-terminal domain of DYRK1A and controls its nuclear localization ([PMID: 21777625](#)), and is required for normal DYRK1A/1B protein levels ([PMID: 30496304](#)).

Consistent with this conserved role, the only characterized *S. pombe* protein-kinase interactor of Dca7, **Ppk15** (SPAC823.03; two-hybrid [PMID: 26771498](#)), carries **both** GO:0004674 (protein Ser/Thr kinase) and GO:0004713 (protein Tyr kinase) activities — the **dual-specificity signature of the DYRK family** (the canonical pombe DYRK, Pom1, is likewise annotated). This is strong circumstantial evidence that fission-yeast Dca7 preserves the ancestral DYRK-scaffold function.

Balance note: human WDR68 has also been reported to bind CUL4-DDB1 ([PMID: 25342745](#): *"binds to several proteins, including... Cullin4-damage-specific DNA-binding protein 1 (CUL4-DDB1)"*), so the "DCAF" designation is not entirely baseless. But no study demonstrates the protein delivering substrates for CRL4-mediated ubiquitination — the defining activity of a genuine substrate receptor.

Finding 4 — The AlphaFold model of Dca7 is a confident WD40 β -propeller consistent with the DCAF7/WDR68 fold

Parsing the deposited AlphaFold model **AF-O74763-F1-model_v6** (435 residues) gave a mean pLDDT of **78.6** (median 92.3), with **73.3%** of residues confident (pLDDT ≥ 70) and **59.3%** very high (≥ 90). The InterPro-defined WD40 propeller core (residues **80–345**) has a mean pLDDT of **89.5** (high confidence), while the N- and C-terminal extensions are low-confidence/flexible.

This matches the experimentally described seven-bladed β -propeller of WDR68 ([PMID: 25342745](#): *"Computer-aided structural analysis suggested that WDR68 forms a seven-bladed β -propeller ring."*) and the observation that regions outside the central WD40 repeats mediate DYRK1A binding ([PMID: 21777625](#)). The structural prediction is fully compatible with the kinase-scaffold model: a stable WD40 propeller core presenting flexible terminal arms that, in the orthologs, engage kinase partners. The fold alone does not distinguish DCAF from scaffold — WD40 propellers serve both roles — so structure neither confirms nor refutes the CRL4 claim; it simply confirms family membership.

Mechanistic Model / Interpretation

The central question is whether the CRL4 substrate-receptor activity, or the kinase-scaffold activity, represents the **core (primary)** molecular function of Dca7. The evidence favors the kinase scaffold.

COMPETING MODELS FOR Dca7 CORE FUNCTION

(A) SEED HYPOTHESIS – CRL4 substrate receptor

Pcu4/Cul4 — Ddb1 — [Dca7?] — (substrate) → ubiquitination

^

| requires WDxR DDB1-binding motif X ABSENT

| requires interaction with Ddb1/Cul4 core X NONE

| requires a demonstrated substrate X NONE (any organism)

SUPPORT: orthology/name only (ISS, GO_REF:0000024)

(B) COMPETING / BETTER-EVIDENCED – kinase scaffold

DYRK/HIPK/MEKK kinase ==bind==> Dca7 (WD40 propeller, catalytically inactive)

|

|

+--> controls kinase stability & levels

+--> controls kinase nuclear localization

|

+--> regulates gene expression / signaling relay

SUPPORT: direct assays across human/mouse/plant/worm orthologs;

pombe interactor Ppk15 is a DYRK-type dual-specificity kinase

The seed hypothesis conflates a **name** ("DCAF7") with a **demonstrated activity**. The DCAF7 name was coined when the human protein was catalogued among Cul4-Ddb1-associated factors, but subsequent functional work redefined the protein's principal biology as kinase scaffolding. The name has since propagated to orthologs — including *S. pombe* Dca7 — via automated and manual orthology transfer, carrying the CRL4 complex annotation with it. In fission yeast, none of the three hallmarks of a genuine substrate receptor is met: (i) no WDxR docking motif, (ii) no interaction with the CRL4 core, and (iii) no demonstrated substrate. By contrast, the scaffold model is supported by a conserved fold, a conserved family function, and a locally relevant kinase interactor (Ppk15).

Importantly, *S. pombe* **does** possess well-characterized, genuinely CRL4-associated DCAFs — Raf1 (PMID: 22319459), Cdt2 (PMID: 17039252), Wdr70 (PMID: 27098497) — each with direct biochemical/phenotypic evidence and (for Raf1) a validated WDxR motif. Dca7 is conspicuously absent from this list, reinforcing that its CRL4 annotation is a carry-over rather than a curated experimental result.

Discriminating criterion	Genuine pombe DCAFs (Raf1, Cdt2, Wdr70)	Dca7
WDxR / DDB1-binding motif	Present (Raf1, verified by mutation)	Absent (0 matches)
Recorded interaction with Ddb1/Cul4 core	Yes (biochemical)	None
Demonstrated substrate	Yes (e.g. Cdt1 for Cdt2; Epe1 for Cdt2 complex)	None
Dedicated CRL4 biochemistry paper	Yes	None
Alternative conserved function	—	Kinase scaffold (DYRK/HIPK/MEKK)

Evidence Base / Evidence Matrix

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Context	Confidence & limitations
PomBase / UniProt O74763	Review/ database	Qualifies	Is the CRL4 annotation experimental?	Only CC term GO:0080008 (Cul4-RING) is ISS/ GO_REF:0000024 (ortholog transfer); no MF/BP terms	<i>S. pombe</i> Dca7	High the annotation is ISS; database level
16949367	Direct assay (biochemistry)	Refutes (sets criterion)	What defines a WD40 DCAF	WD40 DCAFs require a conserved WDxR motif to bind Ddb1	Human	High; established the standard Dca7 f
Motif scan (computational, this study)	Computational	Refutes	Does Dca7 have a WDxR motif?	Zero WDxR matches in 435-aa sequence	<i>S. pombe</i> Dca7	High for absence scans of miss n canon variant
PomBase interactions	Interaction/ database	Refutes / Competing	Does Dca7 bind CRL4 core?	Only interactors are Fft3 (remodeller) and Ppk15 (kinase); no Cul4/Ddb1/Rik1/ Cdt2	<i>S. pombe</i>	Medium high; absence evidence not exhaus
26771498	Interaction (two-hybrid)	Competing	Dca7 kinase partner?	Dca7 interacts with Ppk15, a dual-specificity (Ser/Thr + Tyr) DYRK-type kinase	<i>S. pombe</i>	Medium Y2H, si metho

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Context	Confidence & limitations
20940704	Direct assay	Competing	Family core function	Han11 is a scaffold for HIPK2/MEKK1/DYRK kinase signalling	Human	High
27307198	Direct assay	Competing	Catalytic activity & role	DCAF7 has no catalytic activity; adaptor for DYRK1A/HIPK2	Human	High
21777625	Direct assay / localization	Competing	Kinase binding & localization	WDR68 binds DYRK1A N-terminus; controls nuclear translocation	Mammalian cells	High
30496304	Mutant / knockout	Competing	Functional consequence	WDR68 required for normal DYRK1A/1B protein levels	Mouse/human cell lines	High
25342745	Direct assay / structural	Qualifies (balance)	Does WDR68 bind CUL4-DDB1?	WDR68 forms 7-bladed β -propeller; binds DYRK1A and CUL4-DDB1	Human	Medium; association shown; substrate delivery
AlphaFold AF-074763 (computational, this study)	Structural/computational	Qualifies	Fold of Dca7	Confident WD40 β -propeller (core pLDDT 89.5); flexible termini	<i>S. pombe</i> Dca7	High for fold; for shared both models
23349862		Competing				

Citation (PMID)	Evidence type	Supports / Refutes / Qualifies / Competing	Claim tested	Key finding	Context	Confidence & limitations
	Mutant phenotype		Ortholog in vivo role	Wdr68 is a conserved scaffold in a Ras-Map3k-Wdr68-Dyrk1 relay	Zebrafish/vertebrate	High for scaffold role
22319459	Direct assay / mutant	Qualifies (contrast)	What a real pombe DCAF looks like	Raf1 is a bona fide pombe DCAF with WDxR motif for Rik1/Cul4	<i>S. pombe</i>	High; shows is not analog
17039252	Mutant / biochemistry	Qualifies (contrast)	Genuine CRL4 substrate degradation	Cdt2/Ddb1/Cul4 degrade Cdt1 — real substrate-receptor biology in pombe	<i>S. pombe</i>	High; contrast with D lack of substrate
27098497	Direct assay / mutant	Qualifies (contrast)	Another genuine pombe DCAF	Wdr70 is recruited as part of CRL4(Wdr70), stimulates uH2B, aids resection	<i>S. pombe</i>	High; a WD40 DCAF, unlike Dca7

GO Curation Implications

Lead requiring curator verification. The single cellular-component annotation **GO:0080008 "Cul4-RING E3 ubiquitin ligase complex"** on Dca7 is an **ISS orthology transfer** (GO_REF:0000024) with no supporting *S. pombe* experimental evidence, no WDxR motif, and no recorded interaction with the pombe CRL4 core. It should **not** be treated as a demonstrated core function.

Suggested curation actions (for curator verification):

1. **Retain but flag / down-weight** GO:0080008 as orthology-only (ISS), or reassess whether it meets the local evidence bar at all. Do not promote it to represent the gene's core function, and do not assert a substrate-receptor molecular activity on the basis of this term.
2. **Do not add** a molecular-function term implying substrate-recognition/ubiquitin-ligase substrate-adaptor activity — no evidence supports it in any organism.
3. **Consider the better-supported core function.** The conserved family activity is a **kinase scaffold / adaptor**. If a curator wishes to reflect demonstrated biology, candidate directions include a molecular-function reflecting **protein kinase binding / kinase-scaffold activity** and a biological-process reflecting **regulation of gene expression / signal transduction** — but note these rest on ortholog data plus the local Ppk15 interaction, so ISS-level support at best in *S. pombe*. (Avoid the uninformative "protein binding" as a final recommendation.)
4. **CC "nucleus" (GO:0005634, IBA)** is consistent with both models and with WDR68 nuclear localization; it can be retained.

Net: the evidence supports treating the Cul4-RING CC annotation as **non-core / low-confidence orthology carry-over**, and identifies a competing, better-evidenced kinase-scaffold function that a curator may choose to represent conservatively.

GO term	Current evidence	Recommended action (lead)	Rationale
GO:0080008 Cul4-RING E3 ubiquitin ligase complex (CC)	ISS, GO_REF:0000024	Flag as orthology-only/non-core; do not treat as demonstrated	No WDxR motif, no CRL4-core interaction, no substrate in pombe
GO:0005634 nucleus (CC)	IBA, GO_REF:0000033	Retain	Consistent with WDR68 nuclear localization
(candidate) protein kinase binding / kinase-scaffold (MF)	Not currently annotated	Consider adding at ISS level	Conserved family function; local Ppk15 interaction
(candidate) regulation of gene expression (BP)	Not currently annotated	Consider at ISS level	PomBase descriptor "implicated in gene expression"

Mechanistic Scope

Immediate molecular function being tested: whether Dca7 acts as a substrate-recognition subunit that docks onto Ddb1 within a Cul4 RING E3 ligase and thereby delivers substrates for ubiquitination.

- **Direct gene-product activity (what is actually supported):** Dca7 is a WD40 β -propeller protein (confident AlphaFold model). By strong family conservation it is a catalytically inactive scaffold that binds protein kinases; locally it interacts with the DYRK-type kinase Ppk15 and the chromatin remodeller Fft3.
 - **Not supported as direct activity:** substrate-receptor function within CRL4 — this is inferred solely from the name/orthology and fails the WDxR and core-interaction tests.
 - **Downstream / inferred effects (in orthologs, not to be conflated with core activity):** regulation of DYRK1A/1B protein levels and localization, control of proliferation/differentiation balance, craniofacial and developmental patterning, and (in disease contexts) modulation of RAS/IFN signaling. These are consequences of the scaffold activity, not evidence for a ubiquitin-ligase substrate-receptor role.
-

Conflicts and Alternatives

- **Name-driven database carry-over (principal confound).** "DCAF7" encodes a CRL4 association in its very name, which has propagated to orthologs via ISS. This is the most parsimonious explanation for the pombe annotation and is the classic pattern of ortholog over-annotation.
- **Genuine WDR68–CUL4–DDB1 association exists but is not substrate delivery.** [PMID: 25342745](#) shows human WDR68 can bind CUL4–DDB1. This qualifies (does not refute) the possibility that the family retains some CRL4 contact, but no substrate ubiquitination has ever been attributed to it. The seed hypothesis over-reaches from "can associate" to "is a bona fide substrate receptor."
- **Organism-specific divergence.** Even if human WDR68 touches CUL4–DDB1, *S. pombe* Dca7 shows no such interaction in curated datasets and lacks the WDxR motif; fission-yeast biology may not recapitulate the mammalian association.

- **Bona fide pombe DCAFs provide the contrast case.** Raf1, Cdt2, and Wdr70 are genuine pombe CRL4 substrate receptors with direct evidence; Dca7's absence from this well-studied group is itself informative.

Limitations and Knowledge Gaps

1. **No dedicated Dca7 biochemistry in *S. pombe*.** *Checked:* PomBase interactions and references — all high-throughput. *Why it matters:* the entire CRL4 claim hinges on whether Dca7 physically contacts Ddb1/Pcu4. *Resolution:* co-IP / affinity-MS of tagged Dca7 probing for Ddb1, Pcu4, Rik1, Cdt2.
2. **Is Ppk15 a functional DYRK partner of Dca7?** *Checked:* GO annotations (dual-specificity) and a single Y2H. *Why it matters:* would confirm the conserved scaffold role locally. *Resolution:* co-IP, kinase-level/localization changes in *dca7Δ*, epistasis.
3. **Does Dca7 deletion cause any CRL4-pathway phenotype** (heterochromatin, Cdt1/Spd1 degradation, DSB resection)? *Checked:* deletion is viable; no CRL4 phenotype recorded. *Why it matters:* a genuine substrate-receptor deletion should perturb a defined CRL4 output. *Resolution:* targeted phenotyping of *dca7Δ* against known CRL4 readouts.
4. **Non-canonical DDB1 docking.** *Checked:* WDxR scan (absent). *Why it matters:* some DCAFs use non-WDxR helix-loop-helix docking; absence of WDxR is not absolute proof. *Resolution:* structural docking / cross-linking against Ddb1.

Additional analysis limitations: "No recorded interaction" reflects curated PomBase datasets, which are not exhaustive; a negative in databases is not a definitive experimental negative. The Ppk15 link rests on a single two-hybrid observation plus GO-level dual-specificity annotation. No wet-lab experiments were performed here; the analysis integrates public sequence, structure (AlphaFold), orthology, interaction, and literature evidence.

Discriminating Tests

1. **Affinity purification–MS of endogenously tagged Dca7** in *S. pombe*: presence/absence of Ddb1, Pcu4/Cul4, Rik1, Cdt2 would directly settle the CRL4-membership question; enrichment of Ppk15/other kinases and Fft3 would support the scaffold model.

2. **Reciprocal tagging of Ddb1/Pcu4** to test whether Dca7 co-purifies with the CRL4 core under conditions where Raf1/Cdt2 do.
3. **Phenotypic epistasis:** compare *dca7Δ* to *cdt2Δ/ddb1Δ/raf1Δ* for Cdt1 stability, Spd1 degradation/dNTP levels, H3K9me/heterochromatin silencing, and DSB resection. A substrate-receptor role predicts a specific, shared CRL4 defect.
4. **Structure-guided docking / mutagenesis:** test any predicted Ddb1-contact surface; conversely map the Dca7–Ppk15 interface to the propeller-flanking regions used by WDR68 to bind DYRK1A.
5. **Cross-species complementation:** can Dca7 rescue *wdr68/DCAF7* kinase-scaffold phenotypes but not CRL4-substrate phenotypes?

Proposed Follow-up Actions / Curation Leads (require curator verification)

- **Candidate action change:** Reclassify GO:0080008 (Cul4-RING E3 ubiquitin ligase complex) on Dca7 as **non-core / low-confidence orthology transfer (ISS, GO_REF:0000024)**; do not treat it as evidence of a substrate-receptor molecular function. Do not assert an MF implying CRL4 substrate-adaptor activity.
- **Candidate reference + snippet to verify (criterion for real DCAFs):** [PMID: 16949367](#) — *"the interaction of WD40-containing DCAFs with Ddb1 requires a conserved 'WDXR' motif."* Use to justify demoting Dca7 (WDxR absent).
- **Candidate reference + snippet (competing core function):** [PMID: 27307198](#) — *"DCAF7 is an evolutionarily conserved protein with a single WD40 repeat domain and has no catalytic activity."*; [PMID: 20940704](#) — *"identify Han11 as a novel scaffold protein regulating kinase signalling by HIPK2 and MEKK1."*
- **Candidate reference + snippet (balance):** [PMID: 25342745](#) — WDR68 *"binds to several proteins, including... Cullin4-damage-specific DNA-binding protein 1 (CUL4-DDB1)."* Documents association but not substrate delivery.
- **Candidate new/alternative GO directions (ISS-level, for curator judgment):** a molecular-function reflecting **protein-kinase binding / kinase-scaffold activity**, and a biological-process reflecting **regulation of gene expression** — consistent with PomBase's "implicated in gene expression" descriptor and the Ppk15 interaction. Retain **GO:0005634 nucleus (IBA)**.

- **Suggested questions for curators:** Is GO:0080008 desired at all under local evidence standards? Should the gene's primary function annotation reflect the kinase-scaffold role instead? Is there any unpublished pombe co-IP linking Dca7 to Ddb1?
 - **Suggested experiments:** Dca7 affinity-MS for CRL4 subunits vs. kinases; *dca7Δ* phenotyping against canonical CRL4 outputs; Dca7–Ppk15 interaction validation.
-

Report generated from a 3-iteration autonomous investigation. Computational analyses (WDxR motif scan of O74763; AlphaFold AF-O74763-F1-model_v6 pLDDT parsing) were executed during the investigation; all literature attributions derive from verified PubMed abstract snippets recorded in the knowledge state.

Generated by OpenScientist — Scientific Hypothesis Agent for Novel Discovery